

**Question:**

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A.  $2.48 \times 10^{-8}$  cm
- B.  $5.92 \times 10^7$  cm/sec
- C.  $8.66 \times 10^{-6}$  cm
- D.  $3.14 \times 10^{-5}$  cm
- E.  $9.81 \times 10^{-4}$  cm
- F.  $5.27 \times 10^{-9}$  m
- G.  $1.23 \times 10^{-7}$  cm
- H.  $4.14 \times 10^{-10}$  m
- I.  $6.625 \times 10^{-27}$  erg sec
- J.  $0.512 \times 10^6$  eV

**Reasoning + Answer:**

<think>  
Given the de Broglie wavelength formula  $\lambda = h/p$ , where  $\lambda$  is the wavelength,  $h$  is Planck's constant ( $6.626 \times 10^{-34}$  Js), and  $p$  is the momentum of the particle. For an electron with a kinetic energy of 1 eV, we need to calculate its momentum. Using the kinetic energy formula  $E = \frac{1}{2}mv^2$  and the standard formula for kinetic energy  $E = \frac{1}{2}mv^2$ , we can find the velocity  $v$  and then the momentum  $p = mv$ . The mass of an electron is  $m = 9.11 \times 10^{-31}$  kg. The energy of 1 eV is  $1.6 \times 10^{-19}$  J. Solving for  $v$  gives  $v = \sqrt{2E/m} \approx 5.93 \times 10^5$  m/s. Then  $p = mv \approx 5.4 \times 10^{-25}$  kg m/s. Finally,  $\lambda = h/p \approx 1.23 \times 10^{-7}$  m. Answer

